

ECON 300 – Labour Economics

Final Exam Review

Winter 2026

Part I: Labour Supply and Labour Force Participation

1. A worker has $T = 112$ hours per week to allocate between leisure L and work h , where

$$h = T - L.$$

The worker earns a wage of $w = \$18$ per hour and has non-labour income of \$240 per week.

- (a) Write the weekly budget constraint in terms of leisure and income.
 - (b) What is the intercept on the income axis?
 - (c) What is the intercept on the leisure axis?
 - (d) What is the slope of the budget line?
 - (e) If the wage rises to \$24, show how the budget line changes.
2. A worker's utility-maximizing choice initially is $L = 80$ hours of leisure per week at a wage of \$20 per hour. After a wage increase to \$25, the worker chooses $L = 76$.
- (a) Compute hours worked before and after the wage increase.
 - (b) Did labour supply increase or decrease?
 - (c) Explain, using substitution and income effects, why a wage increase may either increase or decrease hours worked.
3. A person will only enter the labour force if the market wage is at least as high as the reservation wage. Suppose the person values 1 hour of leisure at \$14 and faces fixed costs of working equal to \$60 per day. A full workday is 8 hours.
- (a) What is the minimum daily earnings needed to just compensate for the value of leisure forgone?
 - (b) Including fixed costs, what is the minimum daily earnings required to work?
 - (c) What is the corresponding reservation wage per hour?
 - (d) If the market wage is \$21, will the person participate?

Part II: Income Maintenance, Welfare, and Work Incentives

4. Consider an income maintenance program with:

$$G = 700 \quad \text{and} \quad t = 0.40$$

where G is the guaranteed weekly income and t is the benefit reduction rate. The market wage is $w = \$25$ per hour.

- Write disposable income C as a function of hours worked h while the worker is on the program.
 - Compute the effective net wage while receiving benefits.
 - Find the break-even level of earnings where benefits fall to zero.
 - How many hours of work does this break-even level correspond to?
 - Explain how a higher benefit reduction rate affects labour supply incentives.
5. Compare two transfer programs for the same worker:
- Program A: demogrant of \$500 per week
 - Program B: guaranteed income of \$500 with a tax-back rate of 50%

The wage is \$20 per hour.

- Under Program A, write the budget line equation.
- Under Program B, write disposable income while benefits are positive.
- Under Program B, what is the effective net wage?
- Which program creates a substitution effect? Explain.
- Which program is more likely to reduce hours worked? Explain carefully.

Part III: Labour Demand in Competitive Markets

6. A competitive firm has production function

$$Q = 40N - N^2$$

where N is labour. The output price is $P = \$3$, and the wage is $w = \$48$.

- Compute the marginal product of labour.
- Compute the value of the marginal product of labour.
- Find the profit-maximizing level of employment.

- (d) Compute total output at the optimal employment level.
7. A firm has labour demand elasticity of -1.4 . Initially it employs 250 workers at a wage of \$30 per hour. The wage rises by 10%.
- (a) What is the predicted percentage change in employment?
- (b) What is the predicted new level of employment?
- (c) Briefly explain what it means for labour demand to be elastic.
8. A firm uses labour N and capital K . The marginal products are:

$$MPP_N = 20, \quad MPP_K = 10.$$

The wage is $w = 40$, and the rental rate of capital is $r = 15$.

- (a) Compute $\frac{MPP_N}{MPP_K}$.
- (b) Compute $\frac{w}{r}$.
- (c) Is the cost-minimizing condition satisfied?
- (d) Should the firm use relatively more labour or relatively more capital? Explain.

Part IV: Monopsony and Wage Determination

9. Suppose the labour supply to a monopsonist is

$$W = 12 + 0.5N$$

and the marginal revenue product of labour is

$$MRP_N = 60 - 0.5N.$$

- (a) Derive the marginal cost of labour.
- (b) Find the monopsony level of employment.
- (c) Find the wage paid by the monopsonist.
- (d) Find the competitive wage and employment.
- (e) Compare the monopsony and competitive outcomes.
10. In a monopsony model, a binding minimum wage of \$32 is introduced. Labour supply remains

$$W = 12 + 0.5N$$

and labour demand is

$$MRP_N = 60 - 0.5N.$$

- (a) Find the maximum employment the firm can hire at the minimum wage from the supply curve.
- (b) Find the firm's desired employment by setting the relevant marginal cost equal to MRP_N .
- (c) Will employment rise or fall relative to the monopsony outcome from the previous question?
- (d) Explain why a minimum wage can increase employment under monopsony.

Part V: Compensating Wage Differentials

11. Two jobs are identical except for workplace risk. Job S is safe and pays \$28 per hour. Job R is risky and pays \$34 per hour.
 - (a) What is the compensating wage differential for risk?
 - (b) If a worker requires at least \$5 per hour as compensation for risk, which job will the worker choose?
 - (c) If another worker requires at least \$7 per hour as compensation for risk, which job will that worker choose?
 - (d) Explain how worker heterogeneity generates sorting across jobs.
12. A worker is willing to accept a dangerous job only if the annual wage premium is at least \$1,200. Suppose the job raises annual fatality risk by $1/20,000$.
 - (a) Compute the implied value of a statistical life (VSL).
 - (b) Explain what this VSL means economically.

Part VI: Human Capital, Education, and Training

13. Johny is deciding whether to complete a 3-year diploma. The annual costs are:
 - tuition and books: \$2,500
 - forgone earnings: \$18,000

If he completes the diploma, he expects annual earnings to rise from \$30,000 to \$38,000 for 20 years after graduation. The discount rate is 5%.

Use the annuity formula:

$$PV = A \left(\frac{1 - (1 + r)^{-T}}{r} \right).$$

- (a) Compute the total annual cost of education.

- (b) Compute the present value of the 3 years of education costs, assuming they are paid at the end of each year.
 - (c) Compute the annual earnings gain from the diploma.
 - (d) Compute the present value of the benefits at the time of graduation.
 - (e) Discount the benefits back to the present.
 - (f) Should Johnny invest in the diploma? Use the NPV rule.
14. A worker receives general training that raises annual earnings by \$3,000 in any firm. Another worker receives specific training that raises annual earnings by \$4,000 only in the current firm.
- (a) Which type of training is more likely to be financed by the worker? Why?
 - (b) Which type is more likely to be financed by the firm? Why?
 - (c) Explain why the distinction matters for wage-setting during training.

Part VII: Discrimination and Oaxaca Decomposition

15. The estimated earnings equations are:

$$\ln W_m = 1.10 + 0.08S + 0.05EXP$$

$$\ln W_f = 0.90 + 0.07S + 0.04EXP$$

Average characteristics are:

$$\bar{S}_m = 16, \quad E\bar{X}P_m = 12$$

$$\bar{S}_f = 15, \quad E\bar{X}P_f = 10$$

- (a) Compute predicted log earnings for men.
- (b) Compute predicted log earnings for women.
- (c) Compute the total log wage gap.
- (d) Using the male wage structure as the reference, compute the explained portion of the gap:

$$(\bar{X}_m - \bar{X}_f)\hat{\beta}_m$$

- (e) Compute the unexplained portion.
- (f) Interpret the explained and unexplained components.

Part VIII: Immigration

16. A country admits 360,000 immigrants in one year:
- Economic: 180,000
 - Family: 108,000
 - Refugees and humanitarian: 72,000
- (a) Compute the percentage share of each category.
- (b) If economic immigrants rise by 15% next year while the other categories remain unchanged, compute the new total number of immigrants.
- (c) Under the new total, what is the new economic immigrant share?
17. A migrant compares annual net earnings of \$48,000 in Country A with \$54,000 in Country B. The one-time moving cost is \$20,000, the planning horizon is 25 years, and the discount rate is 4%.
- (a) Compute the annual earnings advantage of Country B.
- (b) Compute the present value of that advantage using the annuity formula.
- (c) After subtracting moving cost, should the migrant move?

Part IX: Unemployment Measurement and Dynamics

18. In an economy:
- working-age population = 8,000,000
 - employed = 5,360,000
 - unemployed = 340,000
- (a) Compute the labour force.
- (b) Compute the unemployment rate.
- (c) Compute the labour force participation rate.
- (d) Compute the employment rate.
19. Suppose the labour force is 2,500,000 and the unemployment rate is 8.4%.
- (a) How many workers are unemployed?
- (b) How many workers are employed?

- (c) If 35,000 unemployed workers find jobs and 20,000 employed workers lose jobs and start searching, what is the new number of unemployed workers?
- (d) What is the new unemployment rate?

20. In a region,

$$UR \approx \text{Incidence} \times \text{Duration}.$$

Suppose:

- Region A: incidence = 3%, duration = 2.5 months
 - Region B: incidence = 1.5%, duration = 5 months
- (a) Compute the unemployment rate in each region.
 - (b) Which region has higher incidence?
 - (c) Which region has longer unemployment duration?
 - (d) Explain why the same unemployment rate can correspond to different policy problems.